

Recovering FLS Sampling-Composition Shocks

Entropy calibration using OEWS areas

H-2A Project

Idea

- ▶ The FLS reports one wage for each AEWR region and year.
- ▶ Realized respondents may be distributed unevenly within the region.
- ▶ This composition can move the FLS wage.
- ▶ Recover the smallest geographic reweighting that reproduces it.

Interpretation

An implied sampling-composition shock, not the confidential FLS weights.

Objects

For region r , year t , and OEWS area a :

- ▶ w_{art} : OEWS wage for the AEWR occupations.
- ▶ p_{art} : prior weight from county farm employment.
- ▶ $\bar{w}_{rt}^{\text{FLS}}$: regional FLS wage.
- ▶ q_{art} : calibrated weight.

OEWS areas are the wage-information units. Counties remain the units used to construct commuting zones and dissimilarity clusters.

Wage calibration

$$\min_q \sum_a q_{art} \log \left(\frac{q_{art}}{p_{art}} \right)$$

subject to

$$\sum_a q_{art} = 1, \quad \sum_a q_{art} w_{art} = \bar{w}_{rt}^{\text{FLS}}, \quad q_{art} \geq 0.$$

The solution exponentially tilts the prior:

$$q_{art} \propto p_{art} \exp(\lambda_{rt} w_{art}).$$

One wage moment does not identify literal sampling weights. KL divergence selects the closest implied distribution.

Regularization path

Let

$$\bar{w}_{rt}^0 = \sum_a p_{art} w_{art}.$$

Balance to

$$\bar{w}_{rt}^\gamma = (1 - \gamma) \bar{w}_{rt}^0 + \gamma \bar{w}_{rt}^{\text{FLS}}, \quad \gamma \in \{0, .25, .5, .75, 1\}.$$

- ▶ $\gamma = 0$: employment prior.
- ▶ $\gamma = 1$: exact FLS match when feasible.
- ▶ Intermediate values show sensitivity to the composition assumption.

Dissimilarity IV

For target unit k , let $D(k)$ be its dissimilar donor cluster:

$$Z_{kt}^{\gamma} = \frac{\sum_{a \in D(k)} q_{ar,t-1}^{\gamma} w_{ar,t-1}}{\sum_{a \in D(k)} q_{ar,t-1}^{\gamma}}.$$

- ▶ Drop an entire OEWS area if it touches the target cluster.
- ▶ Renormalize the remaining donor mass.
- ▶ Estimate levels with unit and year FE, or changes with year FE.

Why add moments?

One wage restriction can be satisfied by many geographic weight vectors.

Additional moments can:

- ▶ reduce dependence on the prior;
- ▶ connect weights to known FLS sampling dimensions;
- ▶ distinguish composition from a common wage shock;
- ▶ provide overidentifying checks.

Requirement

Each moment needs both a regional FLS target and an area-level analogue.

Feasible public moments

FLS target	Area analogue	Treatment
150+ day worker share	Census of Agriculture Table 7	Soft balance
Quarterly worker shares	QCEW monthly employment	Soft balance
Field and livestock wages	OEWS occupation wages	Sensitivity
Hours per worker	ACS PUMS usual hours	Validation

The first two moments most directly describe worker composition over the year.

Worker-duration moment

FLS target:

$$d_{rt}^{\text{FLS}} = \frac{\sum_s N_{rts}^{150+}}{\sum_s N_{rts}^{\text{all}}}, \quad s \in \{Jan, Apr, Jul, Oct\}.$$

Area analogue:

1. Download county worker counts from Census of Agriculture Table 7.
2. Use 2007, 2012, 2017, and 2022.
3. Aggregate counts to OEWS areas.
4. Interpolate area shares between census years.

Use a soft constraint: the Census counts annual workers, while the FLS uses reference weeks.

Seasonal-employment moment

FLS target:

$$s_{rtm}^{\text{FLS}} = \frac{N_{rtm}^{\text{FLS}}}{\sum_h N_{rth}^{\text{FLS}}}.$$

Area analogue:

1. Download quarterly QCEW files.
2. Keep January, April, July, and October employment.
3. Use private NAICS 111 and 112; exclude agricultural services.
4. Aggregate counties to OEWS areas and form within-year shares.

Balance three quarterly shares. Use a soft penalty because QCEW omits many workers on small farms.

Other candidates

Field and livestock wages

- ▶ FLS regional targets already exist in `fls_region.parquet`.
- ▶ OEWS cores are SOC 45-2092 and 45-2093.
- ▶ Other AEWR occupations span both farm types, so the crosswalk is imperfect. Use only as a sensitivity check.

Hours per worker

- ▶ Use ACS PUMS UHRSWORK for wage workers in NAICS 111/112.
- ▶ PUMA geography and residence-based measurement make this a validation moment, not an exact constraint.

Prior variables are not balancing moments

Useful predictors without public regional FLS targets include:

- ▶ peak-worker size bins;
- ▶ gross-sales class;
- ▶ cropland acres and crop mix;
- ▶ livestock inventories and farm type.

Use them to improve p_{art} or for national validation. Do not impose a balance restriction without a corresponding realized regional FLS target.

Soft multi-moment calibration

For area variables x_{ja} and FLS targets $\bar{x}_j^{\text{FLS}}$:

$$\min_q \sum_a q_a \log(q_a/p_a) + \sum_j \frac{\rho_j}{2} \left(\frac{\sum_a q_a x_{ja} - \bar{x}_j^{\text{FLS}}}{s_j} \right)^2.$$

- ▶ s_j standardizes the moments.
- ▶ ρ_j controls how strongly each target is imposed.
- ▶ Report imbalance, KL divergence, effective area count, and maximum weight.

Recommended sequence

1. Combined wage only: current benchmark.
2. Add the 150-day worker share.
3. Add quarterly worker shares.
4. Test separate wage proxies.
5. Use hours as validation.

Compare weights, support, and first-stage strength after each block.

Sources: FLS, Census of Agriculture, QCEW, and ACS PUMS.